CAPSTONE PROJECT · MARKET RESEARCH & CONTEXT

Pomegranate Procurement in Maharashtra

Market research and context for a data-driven procurement intelligence system

01 Market & value chain

02 Procurement landscape

03 Players & challenges

04 Needs, gap & opportunity

Context deck for KPMG stakeholders · November 2026

Maharashtra anchors India’s pomegranate supply

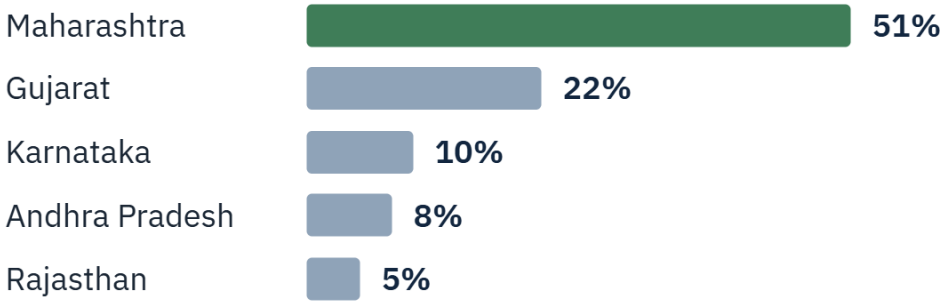
2.9 MMT
Production in 2023-24, up from 0.7 MMT in 2010-11

51%
Maharashtra’s share of national production (54% of area), TE 2023-24

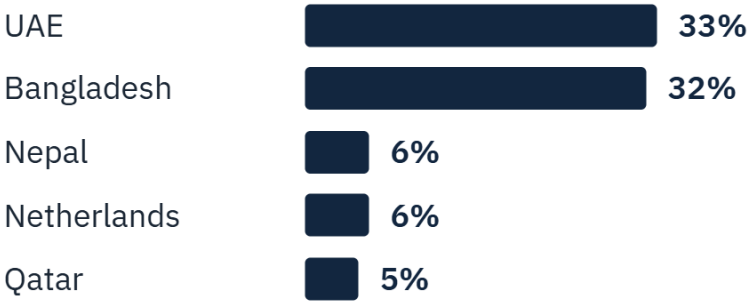
US$69M
Export value in 2023-24, up from US$15.6M in 2010-11

<3%
Share of production exported: the domestic market absorbs most supply

Share of national production, TE 2023-24



Share of India’s exports, TE 2022-23



Source: ICRIER, Strategies to Boost India’s Pomegranate Exports (2026), icrier.org/pdf/Strategies-To-Boost-India-s-Pomegranate-Exports.pdf. TE = triennium ending; shares read from the report’s Figures 2.3 and 3.4.

Value chain: where price and visibility leak

1 Farmer → 2 Aggregator / APMC mandi → 3 Procurement / Agri-Tech firm → 4 Processing, distribution & export → 5 Buyer

Where the chain is inefficient

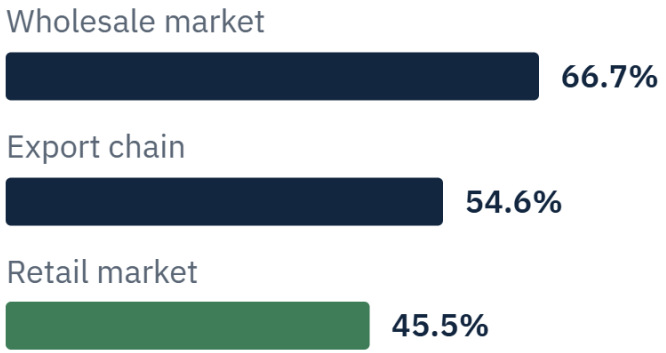
Farm-gate sales. Growers sell through pre-harvest contractors and village aggregators.

Thin grading capacity. Primary collection and grading points are limited before mandi or export.

Cold-chain gaps. Pre-cooling and cold storage are limited ahead of export.

Value split. In the export chain, importers and retailers capture 31.8% of retail value.

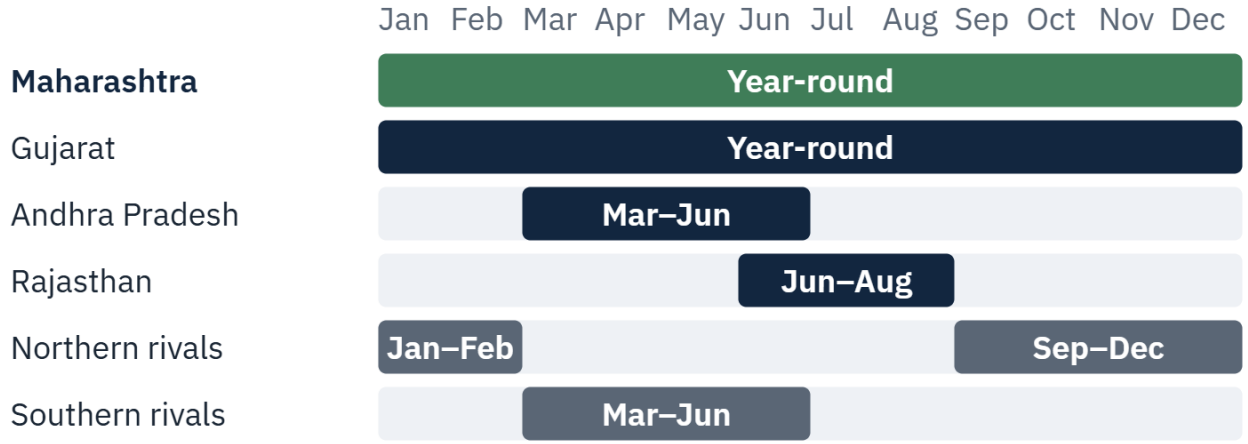
Farmer share of final value



Source: ICRIER, Strategies to Boost India's Pomegranate Exports (2026), Sections 2.5–2.6, icrier.org/pdf/Strategies-To-Boost-India-s-Pomegranate-Exports.pdf. The report's base differs by channel.

Seasonal supply and fragmented mandis drive price swings

Supply windows by origin



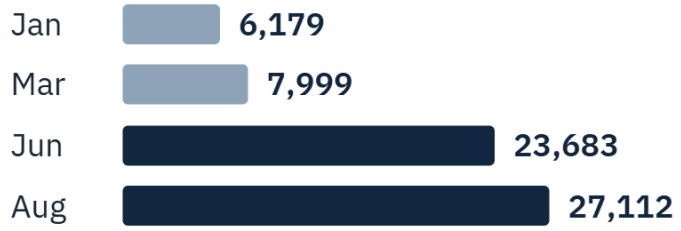
Northern: Iran, Turkey, Egypt, Spain, USA. Southern: Argentina, Peru, South Africa.

~250 g Bhagwa vs 350 g+ preferred

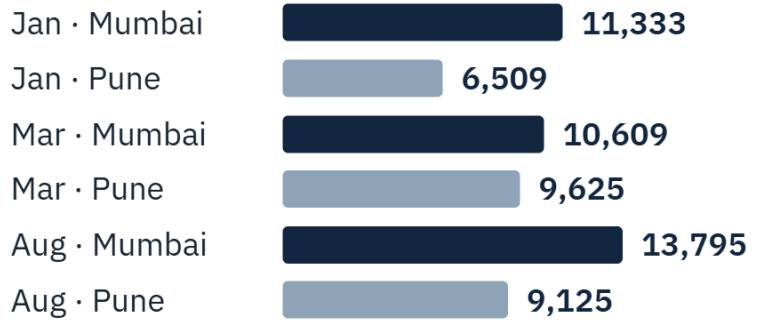
10–12% aril rejection vs 5–7% whole fruit

60–70% of normal output after rains

Pune arrivals, 2024 (MT)



Mandi price, 2024 (₹/qtl)



Sources: ICRIER (2026), icrier.org/pdf/Strategies-To-Boost-India-s-Pomegranate-Exports.pdf · Madrewar et al. (2025), doi.org/10.59890/ijbmp.v3i6.120, Tables 1 and 3 (sampled months only) · FreshPlaza, 2 Feb 2026, freshplaza.com/europe/article/9806856

Peers scale sourcing; predictive sourcing is not yet evident

Player	Sourcing model	Technology use	Farmer network	Market linkage
AgroStar / Kimaye	Farmers sell to Kimaye; packhouse in Baramati	AgroStar advisory app; packhouse quality systems	10k+ farmers sell to Kimaye; 5M+ on AgroStar	25+ countries, 70+ retail chains; US pomegranate trial (2023)
Ninjacart	Buys from farmers, sells to retailers within 12 hours	Demand–supply matching; AI inventory and logistics	200+ collection centres, ~1,400 t/day (2022)	B2B to retailers and restaurants; metro focus
DeHaat	App plus 15,000+ micro-entrepreneur centres	AI crop advisory; AgriCentral app	12M+ farmers across 12 states	Output ~80% of revenue (spices, staples); 32 export markets
Kay Bee Exports	Sources directly from its own farms	Not publicly disclosed	Not publicly disclosed	First sea shipment to New York: ~14 t in under 5 weeks (2025)

Sources: corporate.agrostar.in/marketlinkage · global-agriculture.com (INI Farms, 3 Aug 2023) · decentro.tech/blog/agritech-startups (25 Mar 2026) · startuptalky.com/ninjacart-success-story · startupfeed.in/dehaat-agritech-transforming-bharat · pib.gov.in (PRID 2122827, 19 Apr 2025). Company figures are self-reported or secondary; assessment based on public information reviewed.

06 · KEY INDUSTRY CHALLENGES

Seven pressures converge on the sourcing decision

Price volatility

Jan 2024: Mumbai ₹11,333 vs Pune ₹6,509 per quintal

Uncertain availability

Rains cut Maharashtra output to 60–70% of normal last year

Fragmented sourcing

Farm-gate sales via pre-harvest contractors and village aggregators

Weather risk

Weather is the main uncertainty; heat shortens shelf life

Quality consistency

Bhagwa ~250 g vs 350 g+ preferred; aril rejection 10–12%

Logistics

Aril demand needs cold-chain processing within a week

Limited procurement visibility

ICRIER notes scattered, limited cost and price data

The common gap

No forward view of where, when and at what cost supply is available

Sources: ICRIER (2026), icrier.org/pdf/Strategies-To-Boost-India-s-Pomegranate-Exports.pdf · Madrewar et al. (2025), doi.org/10.59890/ijbmp.v3i6.120 · FreshPlaza, 2 Feb 2026, freshplaza.com/europe/article/9806856. “Common gap” is our synthesis.

07 · FARMER EXPECTATIONS

Farmers want fair prices and reliable market access

Transparent pricing

Farmers get 45.5–66.7% of final value, by channel

Predictable demand

Price fluctuation is among the most-cited marketing constraints (Ahmednagar)

Timely payment

Not a top-ranked constraint in a 2019 study; validate with farmers

Market access

Export prices range from ₹32/kg (Bangladesh) to ₹702/kg (Switzerland)

Lower wastage

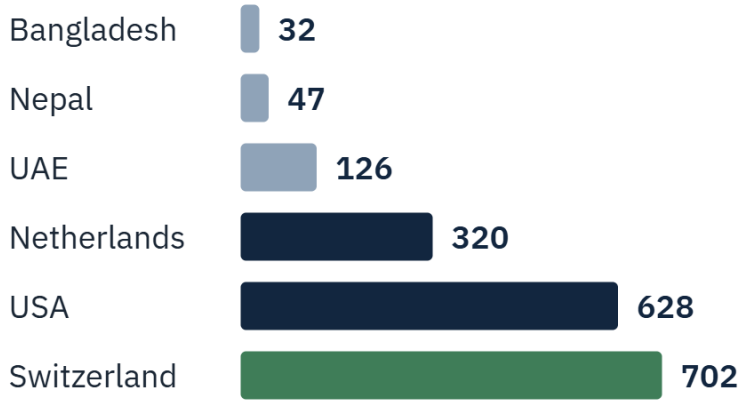
Growers cite missing storage and processing facilities

Price realization

US sea shipment: +20% vs other exports, +35% vs domestic

Export price by destination

₹/kg, TE 2023-24



Implication: transparent pricing and demand signals are the entry ticket for any sourcing partner.

Sources: ICRIER (2026), icrier.org/pdf/Strategies-To-Boost-India-s-Pomegranate-Exports.pdf (Sec. 2.6, Table 3.1, DGFT data) · Jadhav et al. (2019), phytojournal.com/special-issue/2019.v8.i3S.7811. “Implication” is our synthesis.

08 · BUYER & PROCUREMENT EXPECTATIONS

Buyers need reliable supply at a competitive landed cost

✓ Reliable supply

Maharashtra cannot fully meet Gulf demand; orders up 40–45% ahead of Ramzan

Landed cost

Farm-gate ~US$1.7/kg in Gujarat and Rajasthan vs US$2.2/kg for Maharashtra off-season

Consistent quality

350 g+ fruit preferred; Europe applies stricter residue (MRL) rules

Visibility

Peer benchmark: Ninjacart tracks arrivals and prices across ~40 markets

Diversified sourcing

Supply hands off: Maharashtra Apr–Dec, then Rajasthan and Gujarat

Timely delivery

US sea shipment in under five weeks; arils need cold chain within a week

Buyer segments, different priorities

Gulf & South Asia
Volume, price, speed; UAE and Bangladesh take ~65% of exports

EU & US retail
Larger fruit, residue compliance, long shelf life

Aril processors
Cold chain within a week; rejection of 10–12%

Sources: FreshPlaza, 2 Feb 2026, freshplaza.com/europe/article/9806856 · ICRIER (2026), icrier.org/pdf/Strategies-To-Boost-India-s-Pomegranate-Exports.pdf · startuptalky.com/ninjacart-success-story · pib.gov.in (PRID 2122827). Segment framing is our synthesis.

Supply uncertainty vs buyer needs creates the opportunity

Farmer supply uncertainty

- Output swings with weather (60–70% of normal)
- Prices differ by mandi and month
- Sales run through fragmented aggregators
- Demand and price signals arrive late

THE GAP

Buyer procurement needs

- Reliable, predictable supply
- Competitive landed cost
- Consistent quality, timely delivery
- Visibility across origins

Opportunity: a data-driven Procurement Intelligence System

Price Mandi trends

Arrivals Volumes

Weather Crop risk

Logistics Transit, cost

→ **Where, when and how much to source, at what landed cost**

Source for the 60–70% figure: FreshPlaza, 2 Feb 2026, freshplaza.com/europe/article/9806856. Other points synthesize slides 4–8; the system outline is the project's proposed approach.

10 · KEY TAKEAWAYS

Why predictive, optimized procurement creates value

51%	**Maharashtra is the sourcing anchor** About half of India’s production and 54% of area (TE 2023-24)
74%	**Prices differ sharply by market** Mumbai’s Jan 2024 wholesale price was 74% above Pune’s (derived from published data)
60–70%	**Availability cannot be assumed** Maharashtra output ran at 60–70% of normal after last year’s rains
45–67%	**Farmers capture under two-thirds of final value** Retail 45.5%, export chain 54.6%, wholesale 66.7%
Gap	**Peers scale sourcing, not predictive sourcing** No peer reviewed publicly highlights region-level pomegranate sourcing intelligence (our assessment)

Predictive, optimized procurement turns price, arrivals, weather and logistics data into earlier, lower-risk sourcing decisions.

Sources: ICRIER (2026), icrier.org/pdf/Strategies-To-Boost-India-s-Pomegranate-Exports.pdf · Madrewar et al. (2025), doi.org/10.59890/ijbmp.v3i6.120 · FreshPlaza, 2 Feb 2026, freshplaza.com/europe/article/9806856. The 74% gap is our calculation: 11,333 ÷ 6,509 – 1.